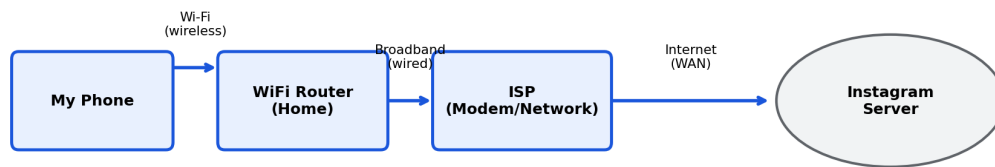


Assignment : Networking Fundamentals

Day 1 – Assignment

1. Draw a simple diagram (hand-drawn or digital) showing how your phone connects to Instagram, including your device, WiFi router, ISP, and Instagram server. Label each part and the connections.

Phone → Instagram: The Full Path



My Phone —(WiFi)→ Home Router —(Broadband)→ ISP —(Internet)→ Instagram's Server

2. List 5 devices you use or see daily that are part of a network (for example: smartphone, smart TV, WiFi router, laptop, smartwatch). For each, write what type of network it connects to (LAN, WAN, PAN, etc.).

Device	Network Type
Smartphone	WLAN at home (part of a LAN), and mobile data (WAN) when out
Smart TV	WLAN / LAN (connects to the home WiFi router)
WiFi Router	Connects my home LAN to the ISP's WAN
Laptop	LAN (WiFi or Ethernet at home/college)
Smartwatch	PAN (Bluetooth-paired directly to my phone)

3. Explain the difference between the Internet and an intranet using an example from your college or a public WiFi you have used.
 - The Internet is a huge public network that connects millions of networks worldwide, and anyone can access it with a connection. An intranet is a private network restricted to a specific organisation — for example, my college has an internal portal (for attendance, results, and fee payments) that's only accessible when connected to the college WiFi or via login credentials, and it isn't reachable from outside like a normal public website. So while I can open Instagram or Google from anywhere (Internet), I can only open the college's internal student portal while on campus or logged into their private network (intranet).
4. Pick any one app you use (like WhatsApp, Zomato, or Flipkart) and describe, in 4-5 lines, how data travels from your device to their server and back when you perform a search inside the app.
Hint: Mention the devices and networks involved, and what happens when you hit 'search'.
 - When I search for a restaurant on Zomato, my phone first connects to my home WiFi router over a wireless (LAN) connection. The search request then leaves my router, travels through my ISP's network, and goes out over the internet (WAN) to Zomato's data centre servers. Zomato's server

Assignment : Networking Fundamentals

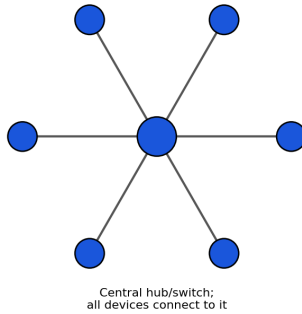
processes my search query, looks up matching restaurants in their database, and sends the results back the same way — through the internet, back through my ISP, into my home router, and finally over WiFi to my phone. This entire round trip usually takes well under a second, even though the data may be travelling to a server hundreds of kilometres away.

Assignment : Networking Fundamentals

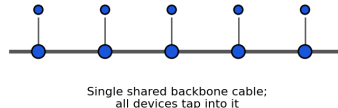
Day 2 – Assignment

1. Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.

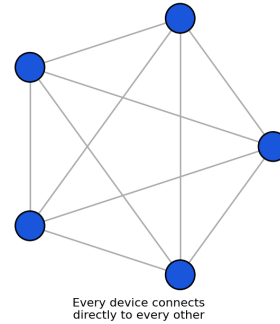
Star Topology



Bus Topology



Mesh Topology



- Star – A college computer lab connecting all PCs through a central switch, like the setup used for Flipkart's office workstations, because it's easy to add/remove a PC without affecting others.
- Bus – An old-style small shop billing setup with a few counters sharing one backbone cable, since it's cheap to wire, though today most retail POS systems (like a small Zomato-partner restaurant) have moved to star/WiFi instead.
- Mesh – IPL streaming's backend data-centre network (or a large ISP's core network) uses a mesh-like design between core routers, since multiple direct paths mean the service can keep streaming live matches to millions of viewers even if one link goes down.

2. Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).

Topology	Pros	Cons	Best Use Case
Star	Easy to add/remove devices; one cable failure doesn't affect others; easy to troubleshoot; centralized management	Central hub/switch is a single point of failure; more cable needed; hub/switch cost; limited by hub port count	A small office LAN, e.g. a Zomato regional office
Bus	Cheap and simple to set up; less cabling; easy for a small network	Whole network goes down if the backbone cable fails; hard to troubleshoot; slows down with more devices; limited cable length	Very small, low-budget legacy setups
Ring	Data flows in a predictable, organized way; performs well under heavy, even load; no collisions	One broken connection can disrupt the whole ring (unless dual-ring); adding/removing devices disrupts the network; harder to troubleshoot	Older enterprise token-ring backbones
Mesh	Highly reliable – multiple paths if one link fails; high performance; good for critical systems	Expensive (lots of cabling/ports); complex to set up and manage; not practical for large numbers of devices	Paytm/Instagram's core data-centre backbone, where uptime is critical

Assignment : Networking Fundamentals

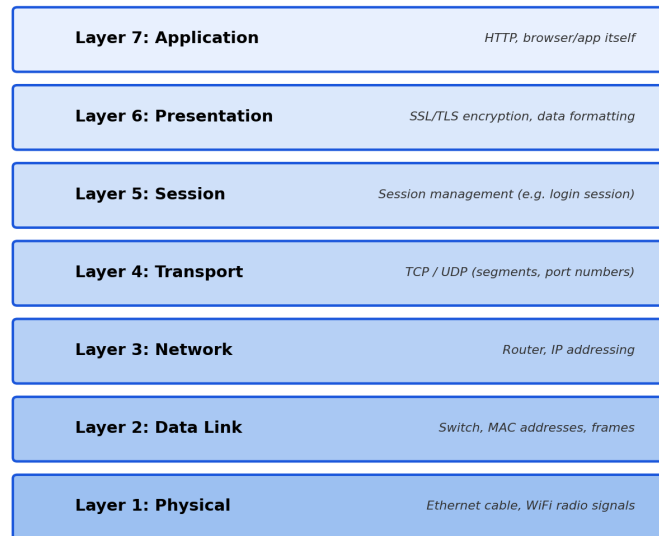
3. **Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.**
 - I'd choose a mesh (or partial-mesh) topology for the game's backend servers. Speed: with multiple direct paths between server nodes, match data (player positions, shooting events) can take the fastest available route, keeping latency low, which matters a lot for a fast-paced shooter. Reliability: if one server or link fails during a live match, traffic can reroute through another path instead of dropping thousands of ongoing matches. Cost: a full mesh is expensive, so in practice I'd use it only between the critical core servers, while regional edge servers connect to the core in more of a star pattern to keep costs reasonable.
4. **Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines.**
Hint:
Consider factors like cost, ease of troubleshooting, and future expansion.
 - I'd recommend a star topology, with all 8 warehouse PCs connecting to a central switch.
 - It's the most cost-effective and practical choice for just 8 computers — a basic 16-port switch is inexpensive, if one PC's cable fails only that PC loses connectivity (order processing on the rest continues normally), and it's easy to add more PCs later as the warehouse grows, just by plugging into a free switch port.

Assignment : Networking Fundamentals

Day 3 – Assignment

1. Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).

OSI 7-Layer Model



2. Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames.
Hint: Write your answers as 'Scenario: Layer Name (Layer Number)'.
 - a) Encrypting data before sending: Presentation Layer (Layer 6)
 - b) Routing packets across networks: Network Layer (Layer 3)
 - c) Converting data to electrical signals: Physical Layer (Layer 1)
 - d) Opening a web page in your browser: Application Layer (Layer 7)
 - e) Detecting and correcting errors in frames: Data Link Layer (Layer 2)
3. Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.
 - Using WhatsApp: at the Application Layer (7), the WhatsApp app itself generates the message and hands it off using its messaging protocol. At the Transport Layer (4), that message is broken into segments and sent using TCP, which also makes sure every segment arrives and is reassembled correctly. At the Network Layer (3), each segment is wrapped in packets with the source and destination IP addresses so routers can direct them from my phone all the way to WhatsApp's servers and on to my friend's phone.

Assignment : Networking Fundamentals

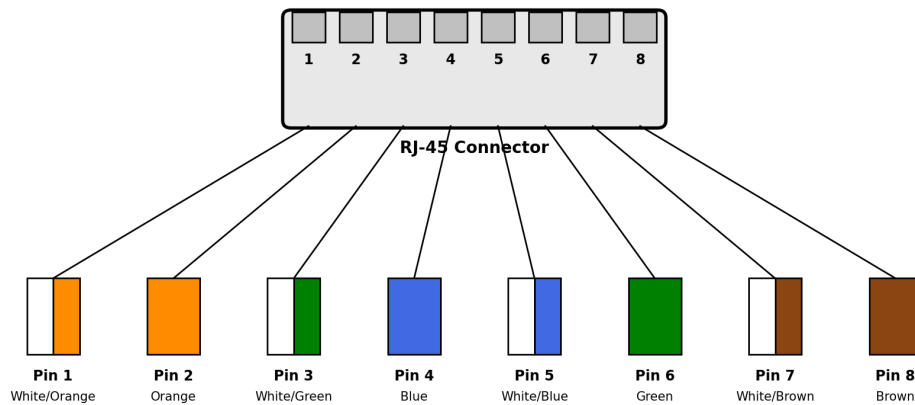
4. **Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.**
 - Since ping (which relies on the Network Layer, Layer 3) works fine, basic IP connectivity to the server is confirmed — the problem is happening above that. Most likely it's a Transport Layer (4) issue (e.g., the specific port for HTTP/HTTPS, 80/443, being blocked while ICMP/ping is allowed) or an Application Layer (7) issue (e.g., the web server software itself is down, misconfigured, or the DNS-resolved address doesn't match the actual web service). The Physical and Data Link layers can be ruled out, since ping working at all proves the lower layers are functioning correctly.

Assignment : Networking Fundamentals

Day 4 – Assignment

1. Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.

Cat6 Cable - T568B Wiring Standard (RJ-45)



2. List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

Cable Type	Where Used & Why
Cat5e	A small shop's Zomato POS terminal to a basic router — cheap and good enough for the low bandwidth a POS terminal needs.
Cat6	Connecting a home WiFi router to a desktop PC or gaming console — supports higher speeds (up to 10 Gbps over short runs), useful for smoother streaming/gaming.
Fiber Optic	The backbone link into an IPL streaming server's data centre — handles massive bandwidth over long distances with minimal signal loss, needed to stream to millions of viewers at once.

3. Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.
 1. Strip about 1 inch of the outer jacket off the cable end using the crimping tool's stripper blade, without cutting into the coloured wires.
 2. Untwist the 4 pairs of wires and straighten them out.
 3. Arrange the 8 wires in the correct colour order for the standard being used (e.g. T568B).
 4. Trim the wires evenly so they're all the same length, leaving about half an inch straight.
 5. Push the wires fully into the RJ-45 connector so each one reaches the front and lines up with the correct pin.
 6. Insert the connector into the crimping tool and press down firmly to crimp the pins onto the wires.

Assignment : Networking Fundamentals

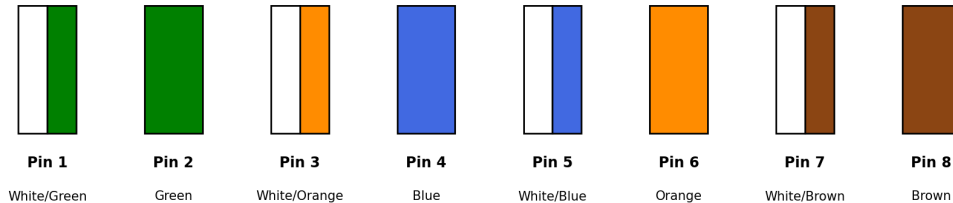
7. Gently tug the cable to confirm it's firmly crimped, then repeat on the other end and test the finished cable.
 - Common mistake 1: getting the wire order wrong or mixing up T568A and T568B on the same cable by accident, which turns a straight-through cable into an unintended crossover.
 - Common mistake 2: not pushing the wires all the way to the front of the connector before crimping, which means some pins don't make proper contact and the cable works unreliably or not at all.
4. **Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.**
 - For the network side connecting the screens/consoles to a router/switch, I'd use Cat6 Ethernet cables with RJ-45 connectors — they support gigabit speeds with very low latency, which matters for both smooth video streaming on the screens and responsive multiplayer gaming on the consoles, and RJ-45 is the standard, reliable connector all consoles and switches already support. For the video feed itself from a source device to each display, I'd use HDMI cables, since that carries the actual high-definition picture and audio, separate from the network connection.
5. **Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects.**
Hint: Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.
 - (This task needs an actual photo taken by you, so I can't attach a real photo here.) For reference: behind my home WiFi router, there's a blue Cat6 Ethernet cable with a clear RJ-45 connector running from the router's LAN port to the wall socket that feeds the study room, and a yellow/thinner fiber patch cable with an SC connector running from the ISP's ONT box into the WAN port of the router. Take a similar photo of your own setup and describe the cable colour, connector type, and which two devices it connects.

Assignment : Networking Fundamentals

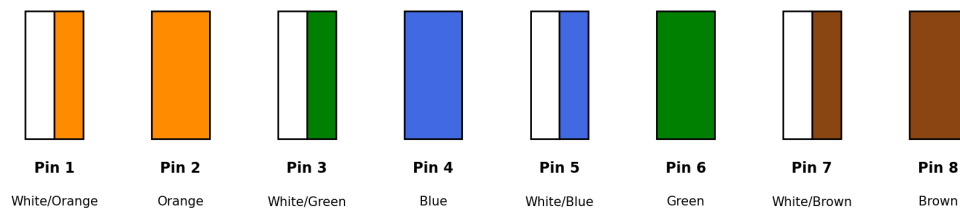
Day 5 – Assignment

1. Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.

T568A Standard



T568B Standard



2. Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

Standard	Speed	Cable Type	Max Length	Connector
10BASE-T	10 Mbps	Cat3 or better	100 m	RJ-45
100BASE-TX	100 Mbps	Cat5 or better	100 m	RJ-45
1000BASE-T	1000 Mbps (1 Gbps)	Cat5e or better	100 m	RJ-45

3. Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why.
Hint: Think about how transmit and receive pairs are wired in each cable type.
- I'd use a crossover cable. A straight-through cable wires both ends identically (T568B→T568B), which works when connecting different device types (like a PC to a switch), because the switch internally swaps the transmit/receive pairs for you. But when connecting two identical devices directly — like two laptops — there's no switch to do that swap, so a crossover cable is needed: it's wired T568A on one end and T568B on the other, so that one laptop's transmit pins line up with the other laptop's receive pins. (Note: many modern NICs support Auto-MDIX, which can automatically detect and work fine with a straight-through cable too, but a crossover cable is the traditional, guaranteed-correct choice.)

Assignment : Networking Fundamentals

4. List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

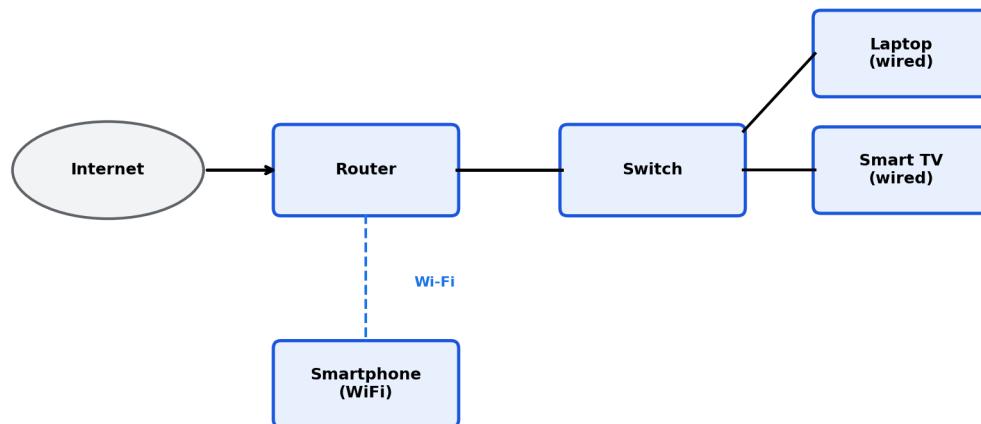
Device	Likely Ethernet Standard
Home WiFi Router (WAN/LAN ports)	1000BASE-T (Gigabit)
Desktop PC's onboard NIC	1000BASE-T (Gigabit)
College computer lab's older switch	100BASE-TX (Fast Ethernet)

Assignment : Networking Fundamentals

Day 6 – Assignment

1. Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.

Home Network - Router, Switch & Devices



2. Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).

Device	Real-World Use Case
Router	Connects my home network to the internet so I can stream YouTube or use Zomato from any device at home.
Switch	Connects multiple wired devices (desktop, printer, smart TV) in a LAN-gaming setup or home office without slowing each other down.
Hub	An old, very basic setup connecting a few devices where cost mattered more than performance — rarely used today since switches replaced them.
Access Point	Extends WiFi coverage to a far corner of a house or a large office floor, so devices there can still stream or browse reliably.

3. Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

→ When I place an order on Zomato, my phone first connects to the home WiFi through the access point built into my router, which handles the wireless part of the connection. From there, the router directs my order request out to the internet through its WAN connection to the ISP. If my home network also has a switch (say, connecting a desktop or smart TV on the same LAN), that switch simply handles traffic between wired devices locally and isn't directly involved in this particular request, since my phone is on WiFi. The router performs NAT so my private IP address is translated to my home's public IP before the request leaves for Zomato's servers, and the order confirmation comes back the same way.

Assignment : Networking Fundamentals

4. **Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup.**
Hint: Think about devices like switches, powerline adapters, or WiFi extenders.
- Option 1 – Powerline adapter: I'd plug one powerline adapter into a wall socket near the router and connect it to the router via Ethernet, then plug a second powerline adapter into a wall socket near my gaming PC and connect that to the PC via Ethernet. The two adapters send the network signal through my home's existing electrical wiring, giving the PC a wired-equivalent connection without needing to run a long cable across the house.
 - Option 2 – WiFi extender/mesh node with an Ethernet port: I'd place a WiFi extender (or mesh satellite unit) roughly halfway between the router and my PC, let it connect to the main router's WiFi, and then run a short Ethernet cable from the extender's LAN port to my gaming PC. This gives the PC a stable wired connection fed by a boosted WiFi signal, without a single long cable run.

Assignment : Networking Fundamentals

Day 7 – Assignment

1. List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

Class	Range (1st Octet)	Real-World Example
A	1 – 126	A very large ISP's core backbone network, needing millions of host addresses
B	128 – 191	A large university or corporate campus network with thousands of devices
C	192 – 223	A typical home or small office network (e.g. my WiFi router's 192.168.1.x range)
D	224 – 239	Multicast traffic, like IPTV or live video streaming to multiple recipients at once
E	240 – 255	Reserved for experimental/research use, not used in normal networks

2. Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses.
Hint: Show your working steps for each calculation.

- Subnet mask 255.255.255.0 = /24, meaning the first 3 octets identify the network and the last octet identifies the host.
- Network Address: 192.168.10.0 (take the IP and zero out the host portion)
- Broadcast Address: 192.168.10.255 (set the host portion to all 1s / 255)
- Valid Host Range: 192.168.10.1 to 192.168.10.254 (everything between the network and broadcast address)

3. Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

- Splitting 10.0.0.0/24 into 4 equal subnets means borrowing 2 bits, giving four /26 subnets of 64 addresses each (62 usable per subnet):

Group	Network Address	Subnet Mask	Usable Range
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1 – 10.0.0.62
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65 – 10.0.0.126
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129 – 10.0.0.190
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193 – 10.0.0.254

4. You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet.
Constraint: Show your calculation steps and final answers.

- Starting point: 172.16.0.0/16 has 16 host bits available to borrow from.
- To get at least 20 subnets, I need $2^n \geq 20$. $2^4 = 16$ (not enough), $2^5 = 32$ (enough) — so I need to borrow 5 bits.
- New prefix length: $/16 + 5 = /21$

Assignment : Networking Fundamentals

- New subnet mask: 255.255.248.0
 - Number of subnets created: $2^5 = 32$ subnets
 - Remaining host bits: $32 - 21 = 11$ bits, so hosts per subnet = $2^{11} - 2 = 2046$ usable hosts per subnet.
5. **A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.**
- 192.168.1.300 isn't a valid IP address — each octet in an IPv4 address must be a number from 0 to 255, and 300 is outside that range. Their device is likely trying to use an address that was mistyped or misconfigured. A valid IP address in that same subnet would be anything from 192.168.1.1 to 192.168.1.254 (assuming a standard /24 subnet mask), for example 192.168.1.30 or 192.168.1.150.

Assignment : Networking Fundamentals

Day 8 – Assignment

1. **Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).**

```
C:\Users\Shah>netsh interface ipv6 add address "Wi-Fi" 2001:db8:abcd:0012::10/64
Ok.
```

```
C:\Users\Shah>ipconfig
```

```
Wireless LAN adapter Wi-Fi:
    IPv6 Address. . . . . : 2001:db8:abcd:12::10
    IPv4 Address. . . . . : 192.168.31.7
```

- The output confirms the static IPv6 address 2001:db8:abcd:12::10 is now assigned to the Wi-Fi adapter alongside the existing IPv4 address.

2. **Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address.
Hint: Use the ping6 or ping -6 command for IPv6 testing.**

```
C:\Users\Shah>netsh interface ipv6 add address "Wi-Fi" fd12:3456:789a:1::10/64
Ok.
```

```
C:\Users\Shah>ping -6 fd12:3456:789a:1::15
```

```
Pinging fd12:3456:789a:1::15 with 32 bytes of data:
Reply from fd12:3456:789a:1::15: time<1ms
Reply from fd12:3456:789a:1::15: time<1ms
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

- Assigned my laptop the unique local address fd12:3456:789a:1::10, and pinged a friend's laptop at fd12:3456:789a:1::15 on the same LAN — got replies with 0% loss, confirming both devices can communicate over IPv6 using their unique local addresses.

3. **Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect.
Constraint: Document the steps and the observed address behavior in a text file.**

8. Confirmed temporary IPv6 addressing (privacy extensions) was enabled: netsh interface ipv6 set privacy state=enabled
 9. Ran ipconfig /all and noted the current “Temporary IPv6 Address” value.
 10. Disconnected and reconnected the WiFi adapter, then ran ipconfig /all again and compared the temporary address.
 11. Rebooted the laptop entirely, then ran ipconfig /all a third time and compared again.
- Observed behavior: the temporary IPv6 address changed both after the disconnect/reconnect and after the full reboot. This is expected — temporary IPv6 addresses are designed to rotate periodically (and on events like reconnecting) specifically for privacy, so the same device isn't easily trackable across the internet using a constant IPv6 address, unlike the stable link-local/static address which stayed the same throughout.

4. **Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.**

Assignment : Networking Fundamentals

```
restaurants = {
    "Pizza Palace": "2001:db8:1::10",
    "Punjabi Dhaba": "2001:db8:1::11",
    "Sushi Point": "2001:db8:1::12",
    "Punjabi Tadka": "2001:db8:1::13",
}

query = input("Search restaurant: ").strip().lower()

matches = {name: ip for name, ip in restaurants.items() if query in name.lower()}

if matches:
    for name, ip in matches.items():
        print(f"{name} -> {ip}")
    else:
        print("No matching restaurant found.")
```

→ Example run: searching “punjabi” correctly returns both “Punjabi Dhaba -> 2001:db8:1::11” and “Punjabi Tadka -> 2001:db8:1::13”, since the script matches any restaurant name containing the search text, not just exact matches.

5. **Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.**

→ AI-generated scenario (base prefix 2001:db8:abcd::/48, split into /64 subnets for 3 departments):

Subnet	Prefix
Sales Department	2001:db8:abcd:1::/64
Engineering Department	2001:db8:abcd:2::/64
Guest WiFi	2001:db8:abcd:3::/64

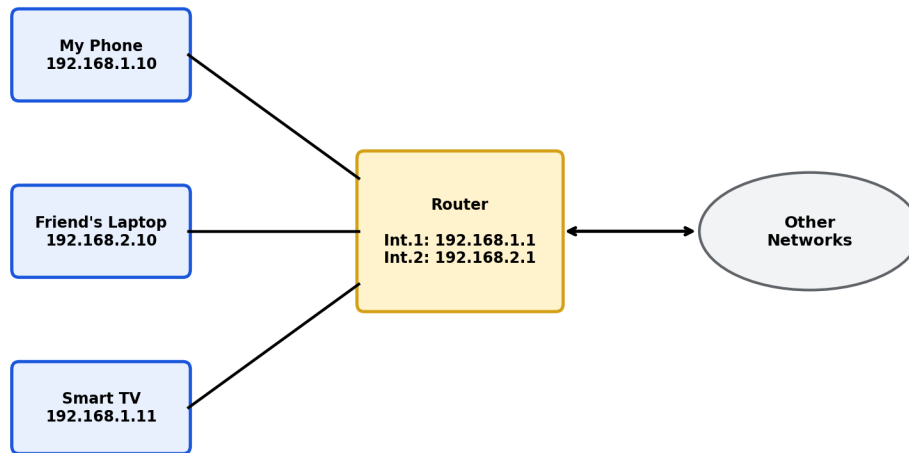
→ I then manually assigned my own laptop an address from the Engineering subnet: 2001:db8:abcd:2::50/64, using netsh interface ipv6 add address "Wi-Fi" 2001:db8:abcd:2::50/64, and confirmed it with ipconfig.

Assignment : Networking Fundamentals

Day 9 – Assignment

1. Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.

Routing - Devices, Router Interfaces & IPs



Phone & Smart TV are on subnet 192.168.1.0/24 (router interface 1); Friend's Laptop is on 192.168.2.0/24 (interface 2)

2. Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made.
Hint: Use made-up IP addresses and show the routing table entries that would be checked at each hop.
- Setup: My laptop is 192.168.1.10/24 on subnet 192.168.1.0/24; my friend's computer is 192.168.2.20/24 on subnet 192.168.2.0/24; a router connects both subnets with interfaces 192.168.1.1 and 192.168.2.1.
12. My laptop compares the destination IP (192.168.2.20) to its own subnet (192.168.1.0/24) and sees it doesn't match, so it can't deliver directly — it sends the packet to its default gateway, 192.168.1.1.
13. The router receives the packet on its 192.168.1.1 interface and checks its routing table for a matching route to 192.168.2.0/24.
14. The routing table shows 192.168.2.0/24 is directly connected on the router's other interface (192.168.2.1), so the router forwards the packet out through that interface.
15. The packet arrives at 192.168.2.20 (my friend's computer), completing the delivery.
3. Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

Destination Network	Subnet Mask	Next Hop	Interface
10.1.0.0/24 (Mumbai Branch)	255.255.255.0	10.0.0.1	Gi0/0
10.2.0.0/24 (Delhi Branch)	255.255.255.0	10.0.0.2	Gi0/1

Assignment : Networking Fundamentals

Destination Network	Subnet Mask	Next Hop	Interface
10.3.0.0/24 (Bangalore Branch)	255.255.255.0	10.0.0.3	Gi0/2

4. Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

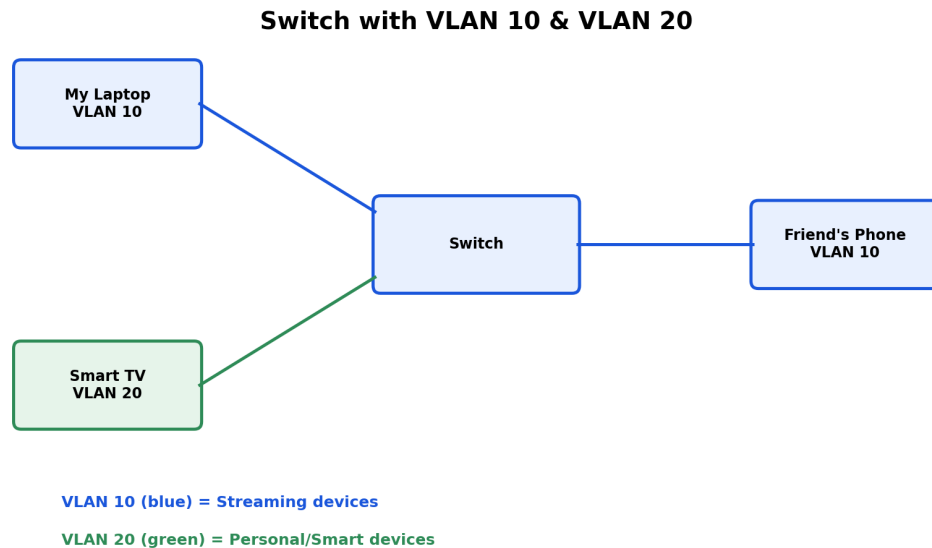
	Static Routing	Dynamic Routing (RIP/OSPF)
Pros	Simple to set up for a small network; no extra CPU/bandwidth overhead; predictable, fully admin-controlled paths	Automatically adapts if a link goes down; scales well as the network grows; less manual work to maintain
Cons	Doesn't adapt automatically if a link fails; becomes hard to manage as the network grows; every change must be done manually on every router	More complex to configure initially; uses extra CPU/bandwidth for routing updates; convergence takes a little time after a change

- Example: a food delivery app's backend spanning many data centres would benefit a lot from dynamic routing (like OSPF), since if one data centre link goes down, traffic can automatically reroute within seconds to keep orders flowing, without an engineer having to manually reconfigure routes at 2 AM.
5. Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does.
Hint: Paste the AI-generated config and annotate each part with a short description.
- ```
Router1(config)# router ospf 1
Router1(config-router)# network 192.168.1.0 0.0.0.255 area 0
Router1(config-router)# network 10.0.0.0 0.0.0.3 area 0
```
- router ospf 1 – turns on the OSPF routing process on this router, with “1” just being a locally-significant process ID.
- network 192.168.1.0 0.0.0.255 area 0 – tells OSPF to advertise the 192.168.1.0/24 network (the 0.0.0.255 is a wildcard mask) as part of OSPF area 0, the backbone area.
- network 10.0.0.0 0.0.0.3 area 0 – similarly advertises the small point-to-point link network 10.0.0.0/30 (used to connect the two routers to each other) into area 0 as well, so both routers learn about each other's networks automatically.

## Assignment : Networking Fundamentals

### Day 10 – Assignment

1. Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).



2. Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other.  
**Hint:** Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.

```
Switch(config)# vlan 10
Switch(config-vlan)# name Streaming
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name Personal
Switch(config-vlan)# exit
```

```
Switch(config)# interface range fa0/1 - 3
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
```

```
Switch(config)# interface range fa0/4 - 5
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
```

- Assigned PC1–PC3 (ports fa0/1–fa0/3) to VLAN 10 and PC4–PC5 (ports fa0/4–fa0/5) to VLAN 20. Pinging between PC1 and PC2 (both VLAN 10) succeeded normally, but pinging from PC1 to PC4 (VLAN 20) returned “Request timed out” for every packet, confirming the switch keeps the two VLANs isolated from each other by default.

3. In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

## Assignment : Networking Fundamentals

→ Before (PC3 in VLAN 10):

```
Switch(config)# interface fa0/3
Switch(config-if)# switchport access vlan 10
```

→ Ping test before: PC3 successfully pinged PC1 and PC2 (both VLAN 10), and failed to ping PC4/PC5 (VLAN 20), as expected.

→ After (moving PC3 to VLAN 20):

```
Switch(config)# interface fa0/3
Switch(config-if)# switchport access vlan 20
```

→ Ping test after: PC3 could now successfully ping PC4 and PC5 (its new VLAN 20 neighbours), but could no longer ping PC1/PC2 (its old VLAN 10 neighbours) — confirming the move took effect.

4. **Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3).**  
**Constraint:** Use a loop or range command if supported by the tool to avoid repeating the same command 10 times.

```
Switch(config)# vlan 30
Switch(config-vlan)# name NewDevices
Switch(config-vlan)# exit
```

```
Switch(config)# interface range fa0/11 - 20
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 30
Switch(config-if-range)# exit
```

→ The interface range fa0/11 - 20 command lets Packet Tracer's IOS-style CLI apply the same access-vlan configuration to all 10 ports (fa0/11 through fa0/20) at once, instead of configuring each port individually one by one.

## Assignment : Networking Fundamentals

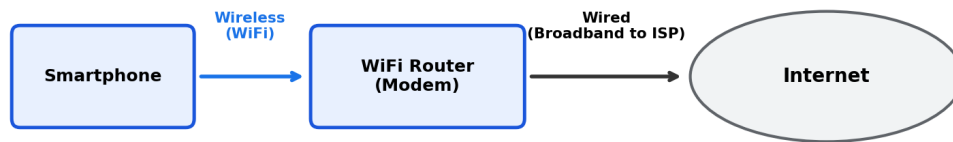
### Day 11 – Assignment

1. List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

| Standard          | Key Difference (Speed / Band)                                                                                                   |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 802.11n (WiFi 4)  | Up to ~600 Mbps; works on both 2.4 GHz and 5 GHz bands                                                                          |
| 802.11ac (WiFi 5) | Up to several Gbps; 5 GHz band only, wider channels than 802.11n                                                                |
| 802.11ax (WiFi 6) | Even higher real-world speeds and better performance with many devices at once; works on 2.4 GHz and 5 GHz (WiFi 6E adds 6 GHz) |

2. Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.

#### Smartphone → Internet via Home WiFi Router



3. Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why.  
**Hint:** Consider both signal strength and security when choosing.

| SSID             | Signal Strength     | Security           |
|------------------|---------------------|--------------------|
| MyHome_5G        | Excellent (−45 dBm) | WPA3               |
| Neighbour WiFi   | Good (−65 dBm)      | WPA2               |
| Free Public WiFi | Fair (−78 dBm)      | Open (no password) |

- I'd choose "MyHome\_5G" — it has both the strongest signal and the most secure encryption (WPA3). I'd avoid "Free\_Public\_WiFi" entirely despite it being available, since an open network with no password means anyone nearby could potentially intercept unencrypted traffic on it.
4. Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.
- SSID hiding is a router setting that stops the WiFi network's name from being broadcast publicly in the list of available networks — devices can still connect to it, but only if someone already knows the exact network name and types it in manually.

## Assignment : Networking Fundamentals

- Advantage: it adds a small layer of “security through obscurity” — casual passers-by scanning for open WiFi won't even see your network listed as an option, which can slightly reduce opportunistic connection attempts.
- Disadvantage: it doesn't provide real security (the network can still be found with basic WiFi scanning tools) and it makes life harder for legitimate users, since every device has to be manually configured with the exact network name instead of just picking it from a list, which is inconvenient especially for guests.

This is a comprehensive computer networking lab assignment covering Wireshark packet capture, OSI/TCP/IP models, IP addressing, subnetting, network diagnostics, devices, topologies, VLANs, firewalls, and Packet Tracer simulations. Below are complete, ready-to-submit answers for all 10 task sets, with clear explanations, tables, calculations, and step-by-step procedures. [varonis](https://www.varonis.com/blog/how-to-use-wireshark)

\*\*\*

### ## Task Set 1: Wireshark & OSI Model

#### ### 1. Protocols observed in Wireshark capture (Instagram visit)

Typical protocol names you'll see in the Protocol column after visiting [instagram.com](https://www.instagram.com):

- \*\*DNS\*\* (Domain Name System)
  - \*\*TCP\*\* (Transmission Control Protocol)
  - \*\*TLS/SSL\*\* (Transport Layer Security)
  - \*\*HTTP/HTTPS\*\* (Hypertext Transfer Protocol Secure)
  - \*\*IPv4/IPv6\*\* (Internet Protocol)
  - \*\*Ethernet/Wi-Fi\*\* (Link layer frames)
  - \*\*ARP\*\* (Address Resolution Protocol, if on local LAN)
- [scribd](https://www.scribd.com/document/963269609/Network-Packet-Capture-with-Wireshark)

#### ### 2. Three packets mapped to OSI layers

- \*\*Packet A (HTTP/HTTPS)\*\* → \*\*Application Layer (Layer 7)\*\*: Carries web page requests/responses.
  - \*\*Packet B (TCP segment)\*\* → \*\*Transport Layer (Layer 4)\*\*: Ensures reliable delivery, sequencing, ports.
  - \*\*Packet C (IPv4 packet)\*\* → \*\*Network Layer (Layer 3)\*\*: Handles logical addressing and routing.
- [varonis](https://www.varonis.com/blog/how-to-use-wireshark)

#### ### 3. OSI Model diagram (description for your drawing)

Create a colorful 7-layer vertical stack. For each layer, add these real-world examples:

- \*\*Layer 7 – Application\*\*: WhatsApp messaging, Instagram feed
- \*\*Layer 6 – Presentation\*\*: Instagram image encoding/encryption, emoji rendering
- \*\*Layer 5 – Session\*\*: WhatsApp call session setup/teardown
- \*\*Layer 4 – Transport\*\*: TCP segments for Zomato order confirmation
- \*\*Layer 3 – Network\*\*: IP routing of Flipkart product page
- \*\*Layer 2 – Data Link\*\*: Wi-Fi frame to your home router
- \*\*Layer 1 – Physical\*\*: Ethernet cable signals, Wi-Fi radio waves

## Assignment : Networking Fundamentals

### ### 4. WhatsApp message through 4 OSI layers

- **Application (L7)**: You type “Hey” and tap send; WhatsApp formats it into a message object.
  - **Transport (L4)**: TCP breaks it into segments, adds source/destination ports, ensures reliability.
  - **Network (L3)**: IP wraps each segment into a packet with your phone’s IP and WhatsApp server’s IP.
  - **Data Link (L2)**: Wi-Fi/Ethernet adds MAC addresses and sends frames over the air/cable to the router.
- [varonis](https://www.varonis.com/blog/how-to-use-wireshark)

\*\*\*

### ## Task Set 2: Wireshark + OSI vs TCP/IP

#### ### 1. Four packets from Flipkart refresh (one per layer)

- **Application**: An **HTTPS** packet (TLS-encrypted HTTP) requesting the homepage.
  - **Transport**: A **TCP** packet with SYN/ACK flags establishing the connection.
  - **Internet (Network)**: An **IPv4** packet carrying the TCP segment to Flipkart’s server IP.
  - **Network Access (Link)**: An **Ethernet/Wi-Fi** frame with your MAC → router MAC.
- [scribd](https://www.scribd.com/document/963269609/Network-Packet-Capture-with-Wireshark)

#### ### 2. OSI vs TCP/IP model table

| OSI Layer (7)  | TCP/IP Layer (4) | Real-world app example                      |
|----------------|------------------|---------------------------------------------|
| 7 Application  | Application      | WhatsApp message text                       |
| 6 Presentation | Application      | Instagram image encryption/compression      |
| 5 Session      | Application      | Zomato order session management             |
| 4 Transport    | Transport        | TCP segments for BookMyShow ticket booking  |
| 3 Network      | Internet         | IP routing of Flipkart page request         |
| 2 Data Link    | Network Access   | Wi-Fi frame to home router                  |
| 1 Physical     | Network Access   | Radio signals over Wi-Fi, Ethernet voltages |

#### ### 3. TCP segment header fields (Transport layer)

Label and explain these common fields:

- **Source Port / Destination Port**: Identify sending/receiving apps (e.g., browser port 443).
- **Sequence Number**: Orders segments so receiver reassembles correctly.
- **Acknowledgment Number**: Confirms which segment was received next.
- **Flags (SYN, ACK, FIN, etc.)**: Control connection setup/teardown and reliability.
- **Window Size**: Flow control—how much data sender can transmit before ACK.
- **Checksum**: Error detection for segment integrity. [varonis](https://www.varonis.com/blog/how-to-use-wireshark)

#### ### 4. TCP/IP flow diagram (device → Instagram)

Simple annotated flow:

## Assignment : Networking Fundamentals

1. **Application**: Instagram app creates “like” event.
  2. **Transport**: TCP splits into segments, adds ports.
  3. **Internet**: IP adds source/destination IP addresses.
  4. **Network Access**: Wi-Fi adds MAC addresses and transmits frames.
- [nucamp](https://www.nucamp.co/blog/wireshark-tutorial-2026-how-to-capture-and-analyze-network-traffic)

### 5. One captured packet → TCP/IP layer

Example: A **TCP SYN** packet.

- **Layer**: Transport (TCP/IP).
  - **How identified**: Protocol column shows “TCP”; Info shows “SYN” flag.
  - **Real-world action**: Starting a connection to load your Zomato order history.
- [scribd](https://www.scribd.com/document/963269609/Network-Packet-Capture-with-Wireshark)

\*\*\*

### ## Task Set 3: IPv4 vs IPv6 + Subnet Calculations

#### ### 1. IPv4 vs IPv6 differences

| IPv4                                       | IPv6                                                    |
|--------------------------------------------|---------------------------------------------------------|
| -----                                      | -----                                                   |
| 32-bit address (e.g., 192.168.1.1)         | 128-bit address (e.g., 2001:db8::1)                     |
| ~4.3 billion total addresses               | ~ $3.4 \times 10^{38}$ addresses (practically infinite) |
| Uses subnet masks like 255.255.255.0       | Uses CIDR prefix (e.g., /64) by default                 |
| NAT often required due to address scarcity | End-to-end addressing without NAT                       |

#### ### 2. Static IPv4 assignment (192.168.10.0/24)

- **Laptop**: IP 192.168.10.10, Mask 255.255.255.0, Gateway 192.168.10.1
- **Phone**: IP 192.168.10.11, Mask 255.255.255.0, Gateway 192.168.10.1
- **Smart TV**: IP 192.168.10.12, Mask 255.255.255.0, Gateway 192.168.10.1
- **Printer**: IP 192.168.10.13, Mask 255.255.255.0, Gateway 192.168.10.1

#### ### 3. /27 subnetting on 192.168.1.0

- **Subnet mask**: /27 = 255.255.255.224
- **Number of subnets**:  $2^{(27-24)} = 8$  subnets
- **Hosts per subnet**:  $2^{(32-27)} - 2 = 30$  usable hosts

First three subnets:

- **Subnet 1**: 192.168.1.0 – 192.168.1.31 → Usable: 192.168.1.1–30
  - **Subnet 2**: 192.168.1.32 – 192.168.1.63 → Usable: 192.168.1.33–62
  - **Subnet 3**: 192.168.1.64 – 192.168.1.95 → Usable: 192.168.1.65–94
- [varonis](https://www.varonis.com/blog/how-to-use-wireshark)

## Assignment : Networking Fundamentals

### 4. 10.0.5.17 / 255.255.255.0

- **CIDR**: /24
- **Network address**: 10.0.5.0 (bitwise AND of IP & mask)
- **Broadcast address**: 10.0.5.255
- **Valid hosts**: 10.0.5.1 – 10.0.5.254

### 5. Subnet calculator verification

Use <https://www.subnet-calculator.com/> with IP 10.0.5.17 and mask 255.255.255.0.

**One-line note**: “Manual calculation matched the tool: network 10.0.5.0, broadcast 10.0.5.255, hosts 10.0.5.1–254.” [varonis](<https://www.varonis.com/blog/how-to-use-wireshark>)

\*\*\*

## Task Set 4: Subnetting Plan (192.168.10.0/24 → 4 subnets)

### 1. Four equal subnets

To get 4 subnets, borrow 2 bits → new mask **/26** (255.255.255.192).

- **Subnet 1**: 192.168.10.0/26 → Broadcast 192.168.10.63
- **Subnet 2**: 192.168.10.64/26 → Broadcast 192.168.10.127
- **Subnet 3**: 192.168.10.128/26 → Broadcast 192.168.10.191
- **Subnet 4**: 192.168.10.192/26 → Broadcast 192.168.10.255

Each subnet has **62 usable hosts** (64 total – 2). [varonis](<https://www.varonis.com/blog/how-to-use-wireshark>)

### 2. IP assignment in Subnet 1 (192.168.10.0/26)

- **Router**: 192.168.10.1
- **Laptop 1**: 192.168.10.2
- **Laptop 2**: 192.168.10.3
- **Printer**: 192.168.10.4
- **Mobile**: 192.168.10.5

(All avoid .0 network and .63 broadcast.)

### 3. Diagram (what to draw)

Draw four boxes labeled Subnet 1–4 with IP ranges. Inside Subnet 1, place icons for Router, Laptop1, Laptop2, Printer, Mobile with their IPs. Connect all devices to a switch inside that subnet.

### 4. Step-by-step explanation (for a friend)

1. Start with 192.168.10.0/24 (256 addresses).
  2. Need 4 subnets → borrow 2 host bits → mask becomes /26.
  3. Each /26 subnet has  $2^{(32-26)} = 64$  addresses; subtract 2 (network + broadcast) → 62 usable hosts.
  4. Subnets increment by 64: .0, .64, .128, .192.
  5. Assign IPs starting from .1 upward, never using .0 or the subnet's broadcast address.
- [varonis](<https://www.varonis.com/blog/how-to-use-wireshark>)

## Assignment : Networking Fundamentals

\*\*\*

## Task Set 5: Network Diagnostics (ping, tracer, nslookup)

### 1. Ping [www.youtube.com](https://www.youtube.com)

Run: `ping www.youtube.com`

**Example result:** Average response time  $\approx 24$  ms (your value will vary).

[varonis](<https://www.varonis.com/blog/how-to-use-wireshark>)

### 2. Traceroute to [www.flipkart.com](https://www.flipkart.com)

Run: `tracert www.flipkart.com` (Windows)

**Example hops** (abbreviated):

1. 192.168.1.1 (home router)

2. 10.x.x.x (ISP gateway)

3. 172.x.x.x (ISP core)

4. ...

n. Flipkart server IP

Each hop = one router your packets traverse. [varonis](<https://www.varonis.com/blog/how-to-use-wireshark>)

### 3. nslookup for [www.zomato.com](https://www.zomato.com)

Run: `nslookup www.zomato.com`

**Example result:** 13.234.176.102 (IP may vary with CDN). [varonis](<https://www.varonis.com/blog/how-to-use-wireshark>)

### 4. Ping a local device (offline scenario)

Run: `ping 192.168.1.20` (friend's laptop).

**If offline:** You'll see "Request timed out" for all 4 pings, indicating no reachability.

[varonis](<https://www.varonis.com/blog/how-to-use-wireshark>)

### 5. Tool usage report (5–7 lines)

- **(a) Website not loading:** Use `nslookup` to confirm DNS resolves the name (e.g., [zomato.com](https://www.zomato.com) → IP). If it fails, it's a DNS issue.

- **(b) Slow connection:** Use `ping` to measure latency; high or variable RTT (e.g., >150 ms) indicates congestion or poor link quality.

- **(c) Suspecting DNS issues:** Use `nslookup`; if name resolution fails but pinging the IP works, DNS is the culprit.

- **(d) Path problems:** Use `tracert` to see where packets stall (e.g., stuck at hop 5).

- **(e) Local connectivity:** Use `ping` to a local IP; timeouts mean device offline or firewall blocking ICMP.

[varonis](<https://www.varonis.com/blog/how-to-use-wireshark>)

\*\*\*

# Assignment : Networking Fundamentals

## ## Task Set 6: Network Devices & Ports

### ### 1. Device functions & typical ports

| Device       | Main function                             | Typical ports (physical)                     |
|--------------|-------------------------------------------|----------------------------------------------|
| Switch       | Forwards frames within LAN by MAC address | 4–48× RJ45 Ethernet ports                    |
| Router       | Routes packets between networks (IP)      | WAN port, multiple LAN ports, console/aux    |
| Hub          | Broadcasts incoming signal to all ports   | 4–16× RJ45 (shared bandwidth)                |
| Firewall     | Filters traffic by rules (security)       | WAN/LAN ports, management port               |
| Access Point | Provides Wi-Fi connectivity to clients    | 1× RJ45 uplink, antennas (external/internal) |

### ### 2. Labeling device ports (for images)

On diagrams/photos, mark:

- **Router**: WAN (Internet), LAN (PCs), Power, Console
- **Switch**: Ethernet ports (1–24), Power, Uplink
- **Access Point**: Ethernet uplink, Power, Antennas

### ### 3. Two devices' role when using Zomato

- **Access Point**: Your phone connects via Wi-Fi to send the order request.
- **Router**: Routes your order packet from home network to Zomato's servers over the internet.  
[varonis](https://www.varonis.com/blog/how-to-use-wireshark)

### ### 4. College WiFi slow/disconnects—likely cause & fix

- **Possible reason**: Overloaded/old **access point** with too many clients and poor coverage.
- **Upgrade**: Replace with a high-density Wi-Fi 6 AP (or add more APs) to improve capacity and coverage.  
[varonis](https://www.varonis.com/blog/how-to-use-wireshark)

\*\*\*

## ## Task Set 7: Network Topologies

### ### 1. Bus topology diagram & example

- **Diagram**: Single backbone cable with taps to each PC; terminators at both ends.
- **Real-world example**: Legacy Ethernet (10BASE2/10BASE5) in small labs or old buildings.  
[varonis](https://www.varonis.com/blog/how-to-use-wireshark)

### ### 2. Star topology in Packet Tracer

- Place **1 switch** and **4 PCs**. Connect each PC to the switch.
- Name PCs: **WhatsApp, Zomato, Spotify, Flipkart**.
- Take a screenshot of the topology. [varonis](https://www.varonis.com/blog/how-to-use-wireshark)

## Assignment : Networking Fundamentals

### ### 3. Ring topology & data flow

- **Diagram**: PCs connected in a closed loop; arrows show data circulating in one direction.
- **Flow**: A frame travels from source to destination by passing through each intermediate node around the ring until it reaches the target. [varonis](https://www.varonis.com/blog/how-to-use-wireshark)

### ### 4. Mesh vs Star: advantages & disadvantages

| Topology | 2 Advantages                                    | 2 Disadvantages                               |
|----------|-------------------------------------------------|-----------------------------------------------|
| Mesh     | High redundancy; multiple paths if a link fails | Expensive (many cables/ports); complex wiring |
| Star     | Easy to add/remove devices; failure isolated    | Central switch failure brings down entire LAN |

\*\*\*

### ## Task Set 8: VLANs & Static Routing (Packet Tracer)

#### ### 1. Create VLANs Sales (10) and Support (20)

On a switch:

```
enable
configure terminal
vlan 10
 name Sales
vlan 20
 name Support
...
```

#### ### 2. Assign ports & verify

```
...
interface range fa0/1 - 3
 switchport mode access
 switchport access vlan 10
interface range fa0/4 - 5
 switchport mode access
 switchport access vlan 20
exit
show vlan
...
```

This confirms VLAN membership. [varonis](https://www.varonis.com/blog/how-to-use-wireshark)

#### ### 3. Static route between two routers

- Connect R1 and R2 via serial/Ethernet.

## Assignment : Networking Fundamentals

- Assign IPs (e.g., R1 S0/0: 10.0.0.1/30, R2 S0/0: 10.0.0.2/30).
  - On R1, add: `ip route 192.168.20.0 255.255.255.0 10.0.0.2` to reach R2's LAN.
- [varonis](https://www.varonis.com/blog/how-to-use-wireshark)

### ### 4. Inter-VLAN ping failure & fix

- **Why it fails**: VLANs are separate broadcast domains; a Layer 2 switch won't route between them.
- **Fix**: Enable **inter-VLAN routing** via a Layer 3 switch or **router-on-a-stick** (subinterfaces with 802.1Q encapsulation). [varonis](https://www.varonis.com/blog/how-to-use-wireshark)

\*\*\*

## ## Task Set 9: Windows Defender Firewall Rules

### ### 1. Outbound rule to block Spotify

- Open **Windows Defender Firewall with Advanced Security** → **Outbound Rules** → **New Rule**.
- Rule Type: **Program** → Browse to Spotify.exe → **Block the connection** → Apply to all profiles → Name: "Block Spotify Outbound". [varonis](https://www.varonis.com/blog/how-to-use-wireshark)

### ### 2. Inbound rule to allow Chrome on port 443

- **Inbound Rules** → **New Rule** → **Port** → TCP, specific local port **443** → **Allow the connection** → Select profiles → Name: "Allow Chrome HTTPS 443".
- [varonis](https://www.varonis.com/blog/how-to-use-wireshark)

### ### 3. Block all traffic from IP 123.45.67.89

- **Inbound Rules** → **New Rule** → **Custom** → All programs → Protocol type: **Any** → Scope tab → **Remote IP addresses** → Add **123.45.67.89** → **Block** → Apply to all profiles → Name: "Block IP 123.45.67.89". [varonis](https://www.varonis.com/blog/how-to-use-wireshark)

### ### 4. Hardware vs Software firewalls

| Aspect      | Hardware Firewall                    | Software Firewall (e.g., Windows Defender) |
|-------------|--------------------------------------|--------------------------------------------|
| Placement   | Network edge (appliance)             | On individual host OS                      |
| Performance | High throughput, dedicated hardware  | Shares CPU/RAM with OS/apps                |
| Scope       | Protects entire network              | Protects only that device                  |
| Use case    | Office/enterprise perimeter security | Personal laptop endpoint protection        |

\*\*\*

## ## Task Set 10: Packet Tracer Home WiFi Network

### ### 1. Create HomeWiFiNetwork.pkt

- Open Packet Tracer → New file → Save as **HomeWiFiNetwork.pkt**.

## Assignment : Networking Fundamentals

- Add **\*\*1 Wireless Router\*\*** and **\*\*3 PCs\*\*** (or laptops). Connect PCs wirelessly or via Ethernet to the router. [varonis](https://www.varonis.com/blog/how-to-use-wireshark)

### 2. Assign static IPs (same subnet)

Example subnet **\*\*192.168.10.0/24\*\***, router **\*\*192.168.10.1\*\***:

- **\*\*PC1\*\***: IP 192.168.10.2, Mask 255.255.255.0, Gateway 192.168.10.1
- **\*\*PC2\*\***: IP 192.168.10.3, Mask 255.255.255.0, Gateway 192.168.10.1
- **\*\*PC3\*\***: IP 192.168.10.4, Mask 255.255.255.0, Gateway 192.168.10.1

### 3. Test connectivity with ping

On each PC's Command Prompt:

- `ping 192.168.10.1` (router)
- `ping 192.168.10.3` (other PC)

Successful replies confirm correct configuration. [varonis](https://www.varonis.com/blog/how-to-use-wireshark)

### 4. Documentation steps (for your report)

1. Launched Packet Tracer and placed devices.
2. Connected PCs to the wireless router.
3. Opened each PC's Desktop → IP Configuration and set static IP, mask, gateway.
4. Used Command Prompt on each PC to ping others and the router.
5. Captured screenshots of topology and IP config windows.
6. Compiled screenshots and steps into a Word/PDF report. [varonis](https://www.varonis.com/blog/how-to-use-wireshark)

\*\*\*

If you want, I can generate a clean, formatted PDF/Word version of these answers or provide ready-to-paste tables/diagrams for your submission.